Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/23/2006
PI Name:	Eckstein, Miguel P Ph.D.		
Project Title:	Quantitative Modeling of Human Saccadic Targeting during Visual Search		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	eckstein@psych.ucsb.edu	**Fax:**	FY 805-893-4303
PI Organization Type:	UNIVERSITY	**Phone:**	805-893-2255
Organization Name:	University of Califonia - Santa Barbara		
PI Address 1:	Vision and Image Understanding Lab		
PI Address 2:	Department of Psychology		
PI Web Page:	http://www.psych.ucsb.edu/		
City:	Santa Barbara	**State:**	CA
Zip Code:	93106	**Congressional District:**	23
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	03/01/2000	**End Date:**	09/01/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A fundamental objective of the Biomedical Research and Countermeasures Program is "to optimize crew safety, well-being, and performance (NASA Research Announcement, p. A-8)" and "to identify, characterize, and mitigate ... environmental ... risks associated with space exploration." Extra vehicular activities (EVA) such as those involved in the construction of the international space station will result in a high workload and high risk environment for human astronauts, thereby increasing the probability of human errors and accidents. Given the advent of high-resolution video-based eye-tracking systems, eye-position interfaces could be potentially used as a powerful and efficient hands free display/control interface and/or to readily access support systems. Such interfaces could mitigate the risks of human errors resulting from the high workload and the fact that the hands are busy with other facets of the task. However, little is known about the human information limits of saccadic targeting. Developing models of human saccadic targeting is therefore imperative so that the eye-position interfaces are well matched to the human visual and occulomotor systems. We hypothesize that a quantitative model based on signal detection theory can be used to predict human saccadic		

	targeting during visual search. To test this hypothesis we propose three specific aims: (1) to measure accuracy of saccadic targeting in elementary visual tasks as a function of basic display properties such as the number of possible locations or elements in the display, target/nontarget discriminability, image noise, and retinal eccentricity; (2) to develop a quantitative model that can reliably predict saccadic targeting accuracy during visual search; and (3) to test and validate the model in two real world search tasks: (a) detecting a lesion in a medical image and (b) locating a target icon among distractor icons on a computer display. The benefits of this research will be a quantitative measures of the information limits of saccadic targeting during visual search and a computational model of saccadic targeting accuracy. These deliverables will help display engineers design advanced interfaces that are well matched to human capabilities.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The potential benefits of a quantitative model for saccadic targeting during search include: a) metrics of image quality to assist engineers in the design of eye-position interfaces (hands free interfaces) for the use by physically disabled people (no longer in command of their limbs and speech) to control different electronic application such as telephone, video-phone, electronic mail, and speech production devices; b) predictions of the eye movement scanning patterns of physicians looking at medical images allowing application of differential image processing techniques (e.g., image compression in progressive transmission of digital medical images) based on predictions about where the eyes are likely to be fixated.
Task Progress:	No progress report this reporting period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)